

Page 116 Equivalence Relation

Str 58 For $a, b \in \mathbb{R}$, define $a \sim b$ iff $a - b \in \mathbb{Z}$.

$\sim$ is reflexive: Let a be any real number.
and since $a - a = 0 \in \mathbb{Z}$.

$\sim$ is symmetric: Let a, b be any real numbers.

If $a \sim b$ then $a - b = k$ is an integer

Therefore $b - a = -(a - b) = -k$ is also an integer

So $b \sim a$.

$\sim$ is Transitive: For any a, b, c in $\mathbb{R}$.

If $a \sim b$ and $b \sim c$ show $a \sim c$.

$a \sim b$ and $b \sim c$ means $a - b = j \in \mathbb{Z}$ and $b - c = k \in \mathbb{Z}$

Thus $a - c = (a - b) + (b - c) = j + k \in \mathbb{Z}$ by closure of $\mathbb{Z}$ under addition.

Therefore $a \sim c$

Since $\sim$ is reflexive, symmetric and transitive, it is an equivalence relation.



$$[0] = \{x \in \mathbb{R} : 0 \sim x\} = \{x \in \mathbb{R} : 0 - x \in \mathbb{Z}\} = \mathbb{Z}$$

$$[.3] = \{.3 + k : k \in \mathbb{Z}\}$$

$$[a] = \{a + k : k \in \mathbb{Z}\}$$

are examples of equivalence classes.

6/10/62 For $(a,b), (x,y)$ in $\mathbb{R}^2$, define $(a,b) \sim (x,y)$ iff $a=x$.

$\sim$ is reflexive: Let (a,b) be any element in $\mathbb{R}^2$
 $(a,b) \sim (a,b)$ since $a=a$.

$\sim$ is symmetric: If $(a,b) \sim (x,y)$ then $a=x$.
So $x=a$ and we have $(x,y) \sim (a,b)$

$\sim$ is Transitive: Let $(a,b) \sim (x,y)$ and $(x,y) \sim (s,t)$
This means $a=x$ and $x=s$. so $a=s$
Therefore $(a,b) \sim (s,t)$

$\sim$ is an equivalence relation on $\mathbb{R}^2$ since it is reflexive, symmetric, and transitive.

Equivalence class examples

$[(0,0)] = \{(0,y) : y \in \mathbb{R}\} =$ points on the y-axis.

$[(1,0)] = \{(1,y) : y \in \mathbb{R}\} =$ points on the vertical line $x=1$

$[(a,b)] = \{(a,y) : y \in \mathbb{R}\} =$ points on vertical line $x=a$.